FloodWay

Predict. Verify. Evacuate safely.

AI flood preparedness and evacuation for Malaysian townships

Sime Darby Property University Hackathon 2026 · Township of Tomorrow · Planet track

Live prototype: floodways.netlify.app

Team information

Team name	**[Team name]**
Team leader	**[Full name]**
University	Universiti Teknologi MARA (UiTM), Shah Alam · Faculty of Computer and Mathematical Sciences
Team members (2–5)	**[Name 1] · [Name 2] · [Name 3] · [Name 4]**
Email	**[Leader email]**
Contact number	**[Leader phone]**

What the team brings

1. **Built and running**
   A live, installable PWA with prediction, photo verification and risk-aware routing.

2. **Improved since March 2025**
   FloodWay began as a UiTM competition project and has been rebuilt and extended since (project report v2.2, Feb 2026).

3. **Right skills**
   Machine-learning deployment, geospatial analysis and full-stack web development.

Fill the highlighted fields before exporting the PDF. Names must match the registration.

Residents cannot answer four questions in a flood

RM6.1bn

flood losses, Dec 2021–Jan 2022 floods (DOSM)

54

deaths in the Dec 2021 floods (police count, 5 Jan 2022)

RM0.6–0.9bn

flood losses a year, 2023–2025 (DOSM)

Four questions residents cannot answer today

1. When will floodwater reach my street?
2. Which road is still passable?
3. Is this flood photo real, or an old one?
4. Which relief centre (PPS) is open and reachable?

Why townships: Bandar Bukit Raja residents reported repeated flash floods since Dec 2021, and SDP lists flood exposure as a key risk. Police and NADMA also keep warning that old flood photos are recirculated.

Target users

Residents and commuters

Families, older residents and drivers in flood-prone townships who must decide when to leave and which way to go.

Township management and committees

Need verified, street-level information to brief residents and run drills.

Responders and authorities

NADMA, Bomba, APM and local councils need verified reports and shelter status.

Sources: DOSM special reports (2021, 2024, 2025); The Vibes (police toll); Sinar Harian (May 2024); SDP Sustainability Report 2023. Full list: Appendix A3.

One flow: predict risk, verify reports, route to safety

1 Predict

A flood probability and risk class for your location, from recent rainfall, forecast and flood history.

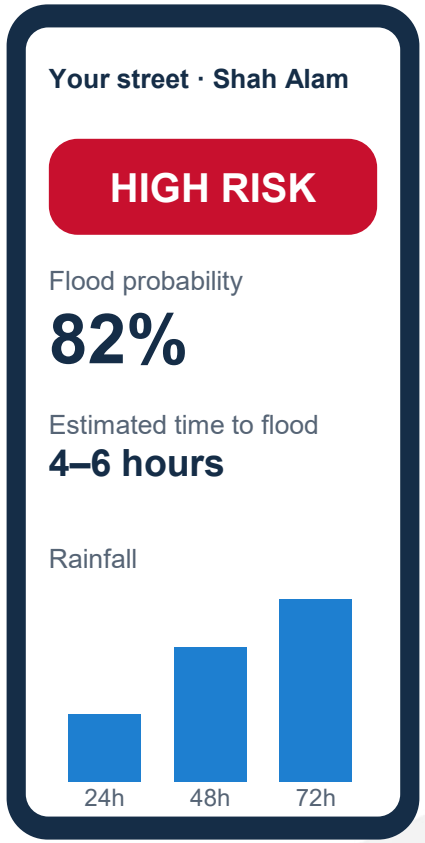
2 Verify

Resident photo reports are checked by AI. Low-confidence reports go to a human moderator before they steer anyone.

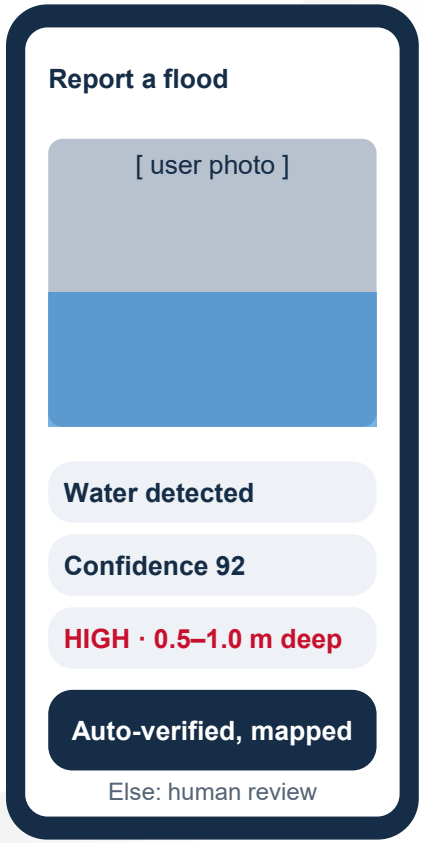
3 Route

Evacuation routes avoid verified flooded roads and end at the nearest reachable relief centre (PPS).

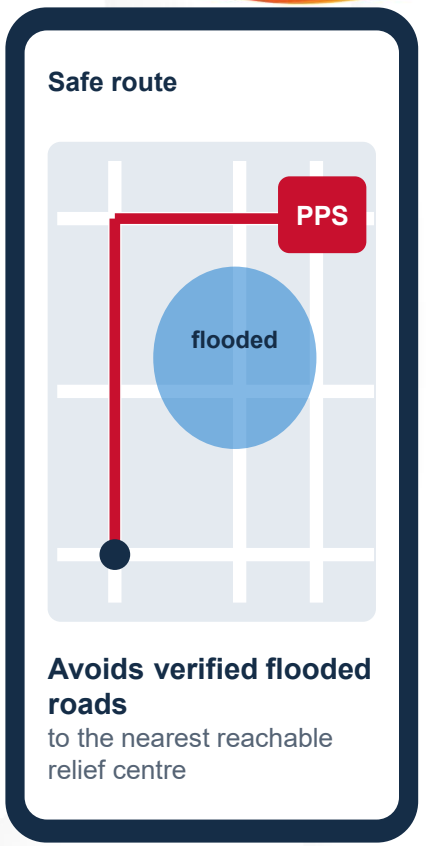
Try it: floodways.netlify.app
Installable PWA, no app-store download



1 Predict



2 Verify



3 Route

Illustrative mockups of the app flow; example values use the prototype's output format. Decision-support only: official warnings remain with JPS and MetMalaysia.

Others cover parts of the chain; we add the last mile

	Waze	Google Maps and Flood Hub	Official Malaysian apps (MyPublicInfoBanjir, MyDIMS)	FloodWay
Flood forecast, 1+ day ahead	Not found	**Yes** · river basins, up to 7 days	**Yes** · JPS bulletins, district and basin level	**Prototype** · district-level model, being validated
Street-level flood timing	Not found	Not found · flash model is 20 km, 24 h	Not found	**Target** · the pilot will test it
Resident flood reports	**Yes** · hazard and closure reports	**Partial** · "flooded road" test, Malaysia unconfirmed	**Yes** · MyDIMS, verified before broadcast	**Yes** · geotagged photo reports
AI check of photo reports	**Partial** · crowd votes only	Not found	Method not described	**Yes** · Gemini vision; accuracy being measured
Route to an open relief centre (PPS)	**Partial** · staff-run crisis pins	Not found	**Yes** · MyDIMS navigation	**Yes** · nearest reachable PPS
Flood-aware route computation	**Partial** · closures only	Not found	Method not described	**Yes** · simulation only; field test in pilot

Where FloodWay is different, if the pilot proves it

Finer than a 20 km forecast cell

Photo reports checked before they steer routes

Flood-aware routing that ends at a relief centre

Checked Oct 2026 against official help pages and news. "Not found" means searched, not proof of absence. Evidence links: Appendix A3.

How it works: three engines on disclosed public data

1 Prediction engine

IN Rainfall over the past 24, 48 and 72 hours, location, flood history of the area, season

HOW Random Forest classifier (scikit-learn), trained on public rainfall and flood-incident data

OUT Flood probability and a risk class (Low, Moderate, High, Critical) with a time-to-flood band

```
{"floodProbability": 0.82,
 "riskLevel": "HIGH",
 "estimatedTimeToFlood": "4-6 hours"}
```

2 Truth filter

IN A geotagged photo from a resident

HOW Gemini Flash vision model scores water, severity and confidence. Above the threshold: auto-mapped. Below: human moderator queue (planned)

OUT A verified flood pin on the map, or a held report

```
{"waterDetected": true,
 "confidence": 92,
 "severity": "HIGH"}
```

3 Risk-aware routing

IN OpenStreetMap roads, verified flood pins, relief centre (PPS) locations

HOW Road graph with risk penalties; verified flooded roads are blocked; OSRM computes the route

OUT Turn-by-turn route to the nearest reachable PPS; a "no safe route" fallback is planned

```
edge weight =
  distance + risk penalty
```

Training data: public sources only, full register in Appendix A1

Dataset 1 Malaysian rainfall 2000–2010: state, district and monthly rainfall patterns

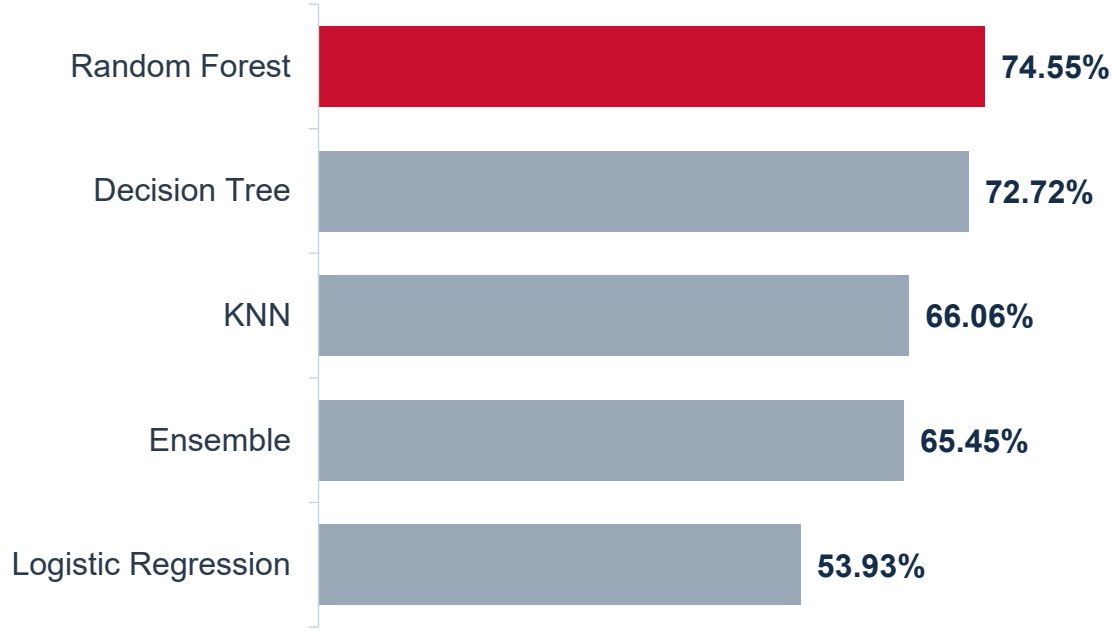
Dataset 2 Malaysian flood incidents 2015–2021 from JPS annual reports: locations, dates, geocoded coordinates, used as supervised labels

Dataset 3 Real-time rainfall forecast: integration ready, used at prediction time

Stack: React, TypeScript, Leaflet and OpenStreetMap on the front end; FastAPI and scikit-learn on Google Cloud Run (planned), Firebase Auth and Firestore, and the Gemini API. Output examples are illustrative.

Proof so far: working prototype, honest baseline

Model comparison: accuracy on an initial hold-out split



Random Forest is selected, but the gap to the Decision Tree is under two points.
We are re-testing on a time-blocked split against a no-flood and a climatology baseline.
Flood share of the test set: **[TBD %]**.

Survey: 100 respondents

ASEAN youth and residents, convenience sample (not representative). Respondents rated AI early warning a priority: **[insert: __ of 100 ...]**

Working prototype

Live PWA with prediction, photo verification and risk-aware routing. Routing tested in synthetic scenarios: simulation, not field-tested.

Measured next, before the pilot

Time-blocked split, PR-AUC and Brier score against baselines, photo-verification false-accept rate, blocked-road routing test.

Model accuracy from the team's initial hold-out split (prototype). Survey is a convenience sample. Method and limits: Appendix A2.

Impact: earlier warnings help, and we will measure ours

Why early warning matters (verified)

~30%

less damage when warning comes 24 hours ahead (Global Commission on Adaptation 2019, via WMO)

8×

higher disaster mortality in countries with limited early warning (UNDRR/WMO 2022)

RM0.6–0.9bn

flood losses a year in 2023–2025 (DOSM)

What we will measure in a township pilot (targets)

Alert lead time	Minutes between a FloodWay alert and the first reported road flooding
Report quality	False-accept and false-reject rate against a labelled photo set
Route safety	Share of test routes that avoid verified flooded roads
Reach	Registered residents and drills completed in the pilot township
Time to safety	Time to reach a PPS in a drill, compared with a map-app route

Value for SDP townships (assumptions to test in the pilot)

- A reusable digital layer on top of wetlands and ponds
- Clearer resident communication and fewer unsafe trips
- One playbook that can repeat across townships. Supports SDG 11 and SDG 13

Sources: Global Commission on Adaptation 2019 via WMO; UNDRR/WMO 2022; DOSM special reports 2023–2025. KPIs are targets, not results. Appendix A3.

Feasibility: a pilot-ready PWA and a one-township plan

Proposed pilot: Bandar Bukit Raja, Klang (subject to SDP agreement). FloodWay adds a digital layer for residents on top of SDP's physical flood mitigation, including its 180 acres of wetlands.

1 Weeks 0–4

Fix and measure

Period-matched rainfall retrain, labelled photo test set, blocked-road routing test

2 Weeks 5–8

Connect

Township contacts, PPS list, JPS and MetMalaysia data access, privacy set-up

3 Months 3–5

Shadow pilot

Monsoon window: alerts reviewed by people before residents see them; resident drills

4 Month 6

Review and scale

Report the KPIs, fix gaps, repeat in the next SDP township

Resources

- Existing PWA and public data
- Google Cloud Run and Firebase (low cost); paid Gemini tier for real resident photos
- Partners to request: SDP township team, local council, JPS, NADMA

Privacy and safety

- Consent, EXIF stripped, coarse location by default, retention limit
- Decision-support only; official warnings stay with JPS and MetMalaysia

Main risks and answers

- Data access: fall back to public and synthetic data
- Model error: show confidence, never a promise
- False reports: threshold plus human review

Pilot township is a proposal to be agreed with SDP. Wetland figure: SDP Sustainability Report 2023. Privacy: PDPA 2010 as amended 2024.

Why our team should win

FloodWay turns a flood warning into a safe decision for one street. Our working PWA predicts risk, checks flood photos with AI before neighbours see them, and routes people to a relief centre.

We tested the market honestly: Waze, Google and MyDIMS cover parts of the chain, so we claim only the last mile and publish how we will measure it.

Our data and assumptions are disclosed, and our pilot plan is ready for a township like Bandar Bukit Raja.

Disclosures

Data Public sources only, listed in Appendix A1. No SDP data used.

AI use Gemini API (Google) in the product; scikit-learn for the models; AI assistants used for research and drafting.

IP Existing FloodWay code is Background IP, licensed to SDP under the T&C.

Links Live prototype: floodways.netlify.app

Code repository and the 5-minute demo video are in the submission.

APPENDIX

Appendix A1: Data sources and assumptions

Item	Source	Licence or limit	Assumption or caveat
Rainfall for model training	Malaysian rainfall 2000–2010, state and district, monthly (public file; confirm exact URL)	Confirm	Monthly resolution; period does not overlap the 2015–2021 labels
Flood labels	JPS annual reports 2015–2021: locations, dates, geocoded points, extracted by the team	Public reports	Only reported incidents; under-reporting likely
Live rainfall forecast	Real-time rainfall forecast API (integration ready; confirm provider)	Confirm	Not part of the accuracy figure
Road network	OpenStreetMap	ODbL, attribution	Road completeness varies
Relief centres (PPS)	NADMA and JKM public information, compiled by the team	No open dataset of locations found	Demo list; live open/closed status needs agency access
Losses and deaths	DOSM special reports; police count (The Vibes)	Public statistics	2022 figure omitted: DOSM revised it
Survey	Team survey, 100 respondents, convenience sample	Own data	Not representative of ASEAN
Simulation	Routing tested in synthetic flood scenarios built by the team	Own data	Not field evidence
AI services	Gemini API (photo check); scikit-learn (prediction); Earth Engine (planned)	Provider terms	Paid Gemini tier for real resident photos

SDP provided no datasets. Items marked "confirm" need the exact URL or licence from the team notebook before submission.

APPENDIX

Appendix A2: Evaluation plan and known limitations

Planned evaluation (time-blocked split)

Metric	Random Forest	No-flood baseline
Accuracy	74.55% (initial split)	**TBD**
Precision / recall / F1	**TBD**	**TBD**
PR-AUC	**TBD**	**TBD**
Brier skill score	**TBD**	n/a
Test-set flood share	**TBD**	n/a

Method

- Train on earlier years, test on later years; hold out whole districts
- Compare models with McNemar test and cluster bootstrap intervals
- Replace monthly rainfall with period-matched satellite or reanalysis rainfall (IMERG, ERA5)

Known limitations

1. Rainfall file and flood labels cover different periods, and rainfall is monthly, so the model is a district-level susceptibility estimate until retrained
2. Accuracy comes from one small initial split; Random Forest and Decision Tree are not clearly different
3. Photo-verification accuracy is not yet measured; vision models can misjudge depth
4. Routing was tested in simulation only; blocked-road behaviour in OSRM needs a field test
5. Prototype was built on Gemini 1.5 Flash, which Google retired in Sep 2025; moving to the current Flash model is a pre-pilot task
6. PPS open/closed status is not live; it needs agency data

Table cells marked TBD are filled from the team's own re-run; no figure here is estimated.

Appendix A3: Sources

SDP University Hackathon page and T&C
https://simedarbyproperty.com/universityhackathon/

SDP Sustainability Report 2023 (flood risk, wetlands)
https://www.simedarbyproperty.com/annualreport2023/download/sdp-sr2...

Sinar Harian: Bukit Raja flash floods, May 2024
https://www.sinarharian.com.my/article/666626/edisi/selangor-kl/pen...

The Vibes: RM6.1bn loss, DOSM
https://www.thevibes.com/articles/business/53117/rm6.1-bil-loss-for...

The Vibes: 54 deaths, 5 Jan 2022
https://www.thevibes.com/articles/news/51355/floods-claim-54-lives-...

DOSM special report, floods 2024
https://www.dosm.gov.my/portal-main/release-content/special-report-...

DOSM special report, floods 2025
https://www.dosm.gov.my/portal-main/release-content/special-report-...

WMO: 24 h warning cuts damage about 30%
https://public.wmo.int/node/25794

WMO/UNDRR 2022: 8× higher mortality
https://wmo.int/node/15497

Waze Help: flood hazard report
https://support.google.com/waze/answer/13809605

Waze Help: road closures
https://support.google.com/waze/answer/13753511

Google Flood Hub: covered locations
https://support.google.com/flood-hub/answer/16508958?hl=en

Google Research: urban flash-flood model
https://research.google/blog/protecting-cities-with-ai-driven-flash...

Malay Mail: NADMA MyDIMS launch
https://www.malaymail.com/news/malaysia/2026/04/24/nadma-launches-m...

Malay Mail: JPS Kelantan forecast, Jan 2026
https://www.malaymail.com/news/malaysia/2026/01/04/heads-up-kelanta...

NST: Waze routes men into floodwater, Nov 2024
https://master.nst.com.my/amp/news/nation/2024/11/1139451/three-res...

Gemini API changelog (1.5 Flash shutdown)
https://ai.google.dev/gemini-api/docs/changelog

Open-Meteo historical API (ERA5)
https://open-meteo.com/en/docs/historical-weather-api

All pages accessed October 2026. Claims we could not source were removed from this deck.